

Week 7

Aim

The aim of this project was to understand the concept of early penetration testing and this seminar was focusing on scanning and finding vulnerabilities in python-based simulation.

Glossary

Penetration Testing – A structured approach to testing and evaluating the security of a system.

Enumeration - Actively searching a system to discover ports and services running on it

Nmap - A network scanning tool used to identify operating systems, services, hosts and open ports.

White box testing – Testing approach where the tester has full access to internal architecture

Black box testing – Testing approach where the tester has no internal knowledge of the system.

Reconnaissance - First stage of penetration testing where information is gathered about a specific target.

Passive reconnaissance – Gathering information without directly interacting with the target system.

Errors and Solutions

When trying to complete the challenge, I had a bit of a learning curve when it came to using the virtual machine as I needed to make my virtual machine accessible to my laptop so that my laptop can perform the enumeration scan and detect its services being hosted. This meant that in my Kali Linux operating system, I was tinkering around with a lot of the network settings to try to get it to become responsive to my script. Eventually the solution was that I needed to change a setting in VirtualBox (virtual machine hosting software) to allow for it to listen to networks. Additionally, I also needed to find out the IP address information too the virtual machine which required me to use some terminal commands as well and to ensure that the IP address was accurate.

Key Takeaways

Penetration testing – After completing this seminar, it reignited a passion of mine since I initially became interested in cybersecurity through penetration testing. Understanding that penetration is also a structured process and is not random hacking was also crucial since a lot of permission and legality is involved.

Nmap - Nmap is a powerful tool, and its importance allows for detection for many services and through using python integration of Nmap I can see how automated process can improve scanning workflows.

Simple Scripts can be powerful – Through writing such a simple script, I was able to identify multiple hosts in my network and gather information on their ports and operating system. Despite the script being so simple, it is so effective and can be used to gather lots of information.